

SUJAY KUMAR MONDAL

Python Backend Developer | Backend Software Engineer | FastAPI | REST APIs | PostgreSQL | Redis

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PROFESSIONAL SUMMARY

Python Backend Developer with 3+ years of experience building scalable RESTful APIs and production-grade backend systems using Python and FastAPI, including enterprise trading and portfolio management platforms for capital markets and OTC derivatives. Experienced in WebSockets, Redis Pub/Sub, PostgreSQL/MySQL, SQL query optimization, JWT/OAuth2, RBAC, microservices, asynchronous programming, gRPC, Kafka, OMS integration, and AI-assisted workflow automation.

CORE TECHNICAL SKILLS

Programming: Python

Frameworks & Libraries: FastAPI, SQLAlchemy, Pydantic, Django

Databases: PostgreSQL, MySQL, SQL, schema design, query optimization, indexing, relational databases

Caching & Messaging: Redis (Streams, Pub/Sub, caching), Apache Kafka

APIs & Authentication: RESTful APIs, JWT, OAuth2, OAuth SSO, Google OAuth, GitHub OAuth, RBAC

Architecture: Microservices, gRPC, WebSockets, event-driven architecture, asynchronous and concurrent programming

Financial Systems: OMS integration, OTC derivatives, interest rate swaps, swaptions, FIX Protocol, capital markets, trading systems

DevOps & Tools: Git, CI/CD, Uvicorn, Gunicorn, Linux, Agile/Scrum

AI / GenAI: AI model API integration, AI-assisted error handling, workflow automation, asynchronous inference processing

PROFESSIONAL EXPERIENCE

System Engineer – Python Backend Developer | Surelia Infosystems Private Limited

Aug 2023 – Present

Portfolio Management & Trading Operations Platform

Tech Stack: Python, FastAPI, WebSockets, Redis, Pydantic

- Build and maintain an enterprise trading platform used by portfolio managers and traders to monitor real-time positions, place orders, and manage complex derivatives workflows.
- Develop FastAPI backend services integrated with WebSocket and Redis Pub/Sub messaging to deliver live portfolio updates, order lifecycle events, and multi-user subscriptions.
- Implement workflow-driven order processing for OTC and structured products, including validation logic and automated booking into external Order Management Systems (OMS).
- Integrate with OMS and internal services for security creation, order booking, allocation, and portfolio data synchronization.
- Design Pydantic-based data models and RESTful APIs for portfolio updates, batch order processing, metadata management, and subscription-based data delivery.
- Implement RBAC-based authentication and role-aware access control to secure multi-user trading operations.
- Introduce AI-assisted error handling and workflow automation to improve reliability and reduce manual intervention in trade processing.

Quantitative Trading & Reference Data Platform (Python)

Tech Stack: Python, REST, gRPC, SQL / Relational Databases, FIX Protocol, Config-Driven Services

- Build and maintain production Python microservices for OTC derivatives reference data, including interest rate swaps and swaptions.
- Extend a modular internal framework for config-driven deployment across development, UAT, and production environments.
- Integrate instrument lifecycle workflows with external portfolio and trading systems through standardized data-access, REST, and gRPC API layers.
- Write unit tests and technical documentation to improve reliability of reference-data and instrument-processing pipelines.

KEY PROJECTS

AI Chatbot — Backend API Service

Tech Stack: FastAPI, PostgreSQL, SQLAlchemy, JWT, Redis, Python

- Designed the backend using asynchronous FastAPI routes, PostgreSQL schema for session-scoped multi-user chat history, and SQLAlchemy ORM models with indexed queries for efficient conversation retrieval.
- Built JWT-based stateless authentication with token refresh and user-scoped access control; integrated external AI model APIs as asynchronous background tasks to decouple inference latency from HTTP responses.
- Structured modular FastAPI routers for authentication, chat, and sessions with Pydantic schema validation, consistent error handling, and RESTful API design.

FastAPI MFA / 2FA Application

Tech Stack: Python, FastAPI, PyOTP, Passlib, SQLite, SQLAlchemy, Jinja2

- Implemented Multi-Factor Authentication (MFA/2FA) using Time-based One-Time Password (TOTP) algorithms in a FastAPI web application.
- Generated QR codes for integration with authenticator applications such as Google Authenticator and Authy.
- Implemented user registration, secure password hashing, login, session management, and protected-route authentication workflows.

EDUCATION

MCA (Master of Computer Applications) | 2021 – 2023 | Brainware University — 88.74%

B.Sc. in Computer Science | 2018 – 2021 | Calcutta University — 65%